Searching for the Anomalous Internal Pair Creation in ^{8}Be

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The purpose of this work is to report the unexpected internal pair creation reaction in ^{8}Be, known as the Hungarian Anomaly, through the interaction ^{7}Li(p,e^{+}e^{-})^{8}Be. Employing a new experimental set up with improved resolution in order to study the anomaly. A detection apparatus consisting of an array of scintillators and a calorimeter block are used to observe the characteristics of the Anomaly and discussion of the X17 Boson. This study aims to examine these detectors, through their characteristics, calibration requirements and procedure and to detect cosmic muons with them and to explore the properties of the experimental methodologies applied and analysis.

I. INTRODUCTION

The investigation of Internal Pair Creation in ^{8}Be is significant in nuclear physics, as it pushes the boundaries of our understanding of nuclear physics and subatomic models. The anomaly observed in the ^{7}Li(p,e^{+}e^{-})^{8}Be reaction, as reported by A.J. Krasznahorkay et al. in the paper *Observation of Anomalous Internal Pair Creation in ^{8}Be: A Possible Signature of a Light, Neutral Boson*, demonstrates a discrepancy with the current theoretical model and could indicate the existence of a new boson.

The purpose of this experiment is to calibrate the detector system to ensure the detector is ready to be used for the e^{+}e^{-} experiment. This is accomplished through background data collection, an α source energy spectrum, and detecting muons based on coincidences with a Plastic Scintillator Detector placed above the detector configuration of interest.

Theoretical Background

In the ^{7}Li(p,e^{+}e^{-})^{8}Be, electron-positron angular correlations were measured for different multipolarity transitions: M1 and M2 transitions from the excited states ($J^{\pi}=1^{+}$, T=1) and ($J^{\pi}=1^{+}$, T=0) to the ground state ($J^{\pi}=0^{+}$, T=0), with energies of 17.6 MeV and 18.15 MeV, respectively.[1] Deviations from the predicted model were observed for large angles. A possible explanation for this deviation is the production of a boson, termed the X17 boson.[2]

In FIG. 1, the results measured by A.J. Krasznahorkay's group are illustrated. The deviation from the theoretical model at an angular distribution of 140 degrees is clearly visible. This plot highlights the purpose of this experiment.

Detector Background

In order to measure the Internal Pair Production of ^{8}Be, a specialized detection system is required. The target reaction, ^{7}Li(p,e^{+}e^{-})^{8}Be, competes with ^{7}Li(p,γ)^{8}Be. In order to ensure the measurement of e^{+}e^{-} events are captured a coincidence is required. Due to energy and momentum conservation, the e^{+} and e^{-} travel in almost the same direction. A time coincidence gate is applied to the system to ensure that only e^{+}e^{-} measurements are collected.

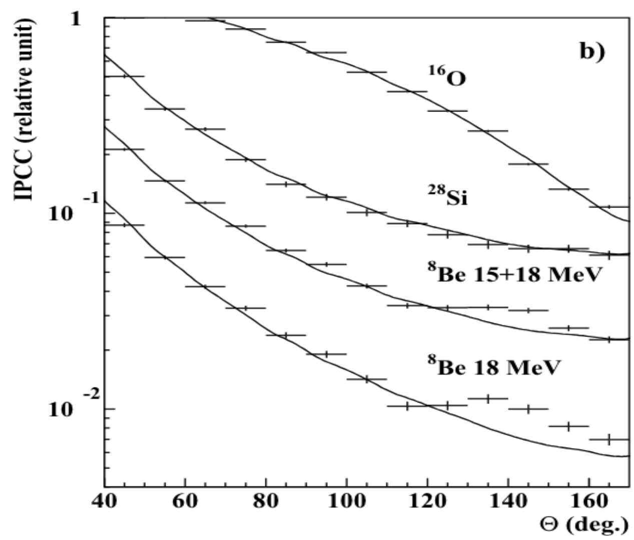

FIG. 1: Electron Positron Angular Distribution [1]

To achieve time coincidence and accurately measure the full energy deposit, a sophisticated detector system is employed. This system includes horizontal and vertical scintillator rectangular bars, coupled with silicon photomultipliers (SiPMs), which are stacked to provide detailed spatial measurements of the events. Additionally, a calorimeter cube is included to capture the total energy deposit. This three scintillator system is titled as a detector telescope. Four such telescopes are arranged to form a clover configuration, enhancing the experiment's solid angle coverage. Multiple clovers are placed in a circular orientation around the target to maximize detection efficiency.

FIG. 2 displays the individual horizontal and vertical scintillator bars with dimensions 50mm x 5mm x 2mm. Ten horizontal and ten vertical bars are used to create an overlap positional grid and the calorimeter with dimensions 100mm x 100mm x 50mm is stacked on top.

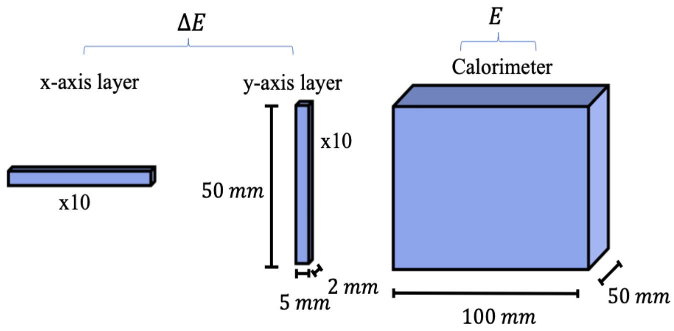

FIG. 2: Detector Diagram

II. METHODOLOGY AND RESULTS

This section is devoted to discuss the activities commenced during the time spent with the experiment and equipment at Istituto Nazionale di Fisica Nucleare (INFN). The aim was to use the detector layout discussed in FIG. 2 to evaluate the detector's performance.

Introduction to Topic and Equipment

A theoretical background on Internal Pair Production in ^{8}Be was provided, along with an overview of the experimental setup, detectors, and expected outcomes. The supervisor, Benito, introduced the results of the previous papers and the implications to the current model understanding of particle physics and the need to reproduce the results. The techniques being implemented at INFN to perform the experiment, specifically focusing on the detector apparatus and showing the beam line. Along with the explanation of the concepts, the plan of action, requirements and deliverables were determined.

Hands on experience with the detector and the vacuum system was carried out in detail and trialed to ensure that the next session, which involved fitting the pipes, verifying the seals to ensure that the detector system can be calibrated after acquiring a source.

Calibration of Scintillator Bars and Calorimeter

With acquisition of an ^{241}Am α radiation source, in order to calibrate the detector system by isolating a single bar with the source in order to observe the positional detection of the radiation, calibrate the gain, the cross talk and functionality of the detector. Once the source and the detector set up was placed inside a vacuum chamber, a leak in the cooling system was discovered due to the flow of fluid in the cooling circuit without the pump being on. As a result the vacuum pump was flooded and immediately operations were halted. Previous samples of calibration data was provided to be analysed in place of our own for practice.

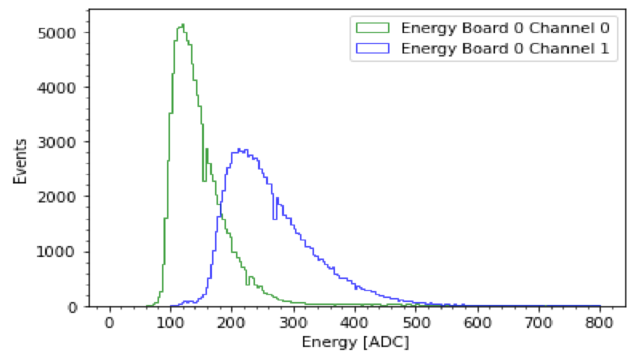

FIG. 3: Left and Right Energy spectra

To determine the position of detection between the beams the expression of the normalisation of the two energy spectra difference is calculated and plotted through the equation.

$$X = \frac{E_1 - E_2}{E_1 + E_2}$$

Here, the value X indicates the relative position of the bar that was exposed to the α source.

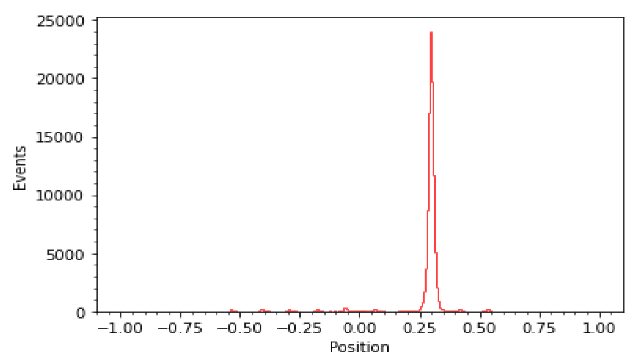

FIG. 4: Position of the Exposed Scintillator Bar

FIG. 4 returns the relative scintillator beam that was exposed to the incident radiation. This peak indicates that the 6th bar was exposed to the radiation. Additionally, no gate in the energy deposited was performed.

A similar procedure was done to observe the energy deposited in the calorimeter and calibration. A dark chamber was used to study the energy deposited in the calorimeter and an α source. Since the setup this time was vacuumless, the source was placed as close as possible to a scintillator bar.

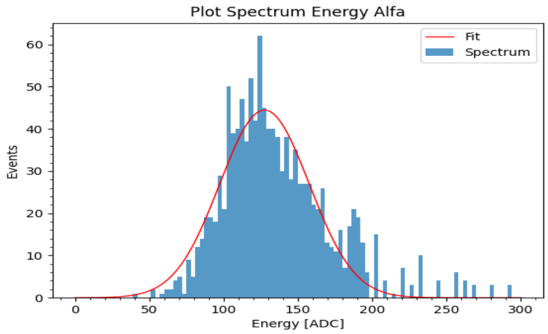

FIG. 5: $\alpha-$Source Calibration

The observed Gaussian distribution, after subtracting the background, is unexpected because ^{241}Am typically exhibits three α peaks. The reduced visibility of these peaks could be attributed to low counts and energy loss of α particles in air before entering the detector.

Calibration, Coincidence and μ Detection

After observing the peak resulting from the α source with the calorimeter. A Plastic Scintillator Detector was placed 19.5 cm above the detector with a thin scintillator for a faster signal generation and a Photomultiplier Tube (PMT). The introduction of this detector is to provide coincidence in order to observe vertically passing muons. The coincidence ensures maximum measurement of muon deposited energy by selecting those that pass through the majority of the detector. Furthermore, calorimeter was oriented parallel (horizontal) or perpendicular (vertical) to the Plastic Scintillator Detector. The measurement period lasted for 3 days in both orientations.

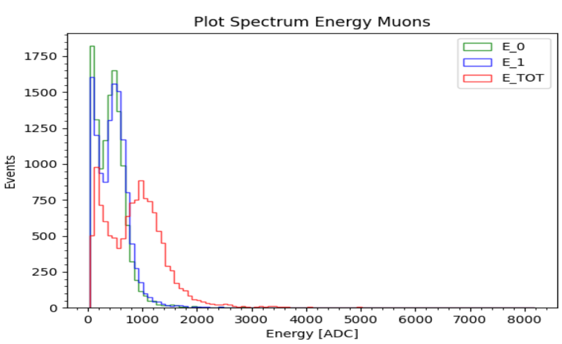

FIG. 6: Muon Detection (horizontal configuration)

In FIG. 6 the green and blue lines indicate the individual detected energy from the channels of the calorimeter. While the red plot indicates the sum of the two. This correlates to the energy deposited in the calorimeter.

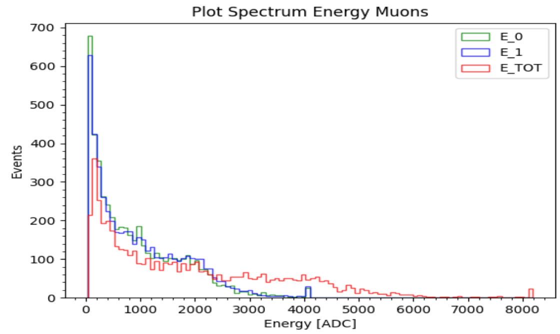

FIG. 7: Muon Detection (vertical configuration)

FIG. 7 illustrates the analysis in vertical configuration. The observed lower count rate is likely influenced by the setup geometry rather than the expected count rate. Additionally, a Landau fit was conducted to illustrate the detector's resolution, although the variation in angles prevented the measurement of the sigma parameter.

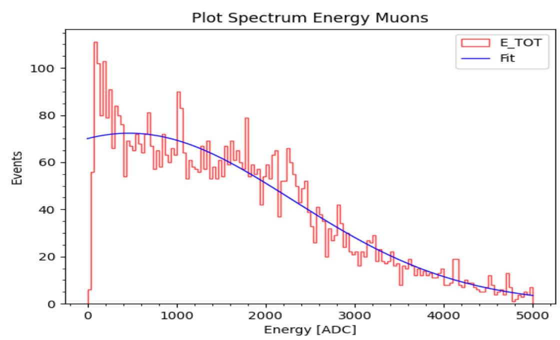

FIG. 8: Landau Fit (horizontal configuration)

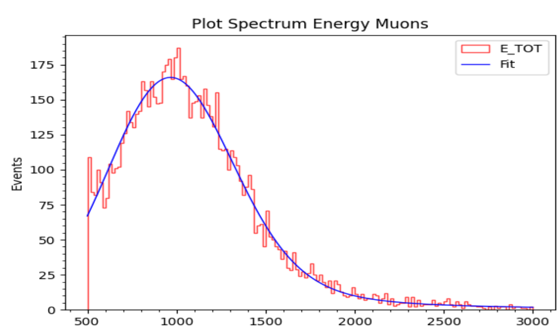

FIG. 9: Landau Fit (vertical configuration)

TABLE I: Acquisition Settings

	calorimeter-A	calorimeter-B	Plastic Scin.
Pre-Trigger (ns)	200	200	96
Polarity	Negative	Negative	Negative
Samples Baseline	256	16	16
Threshold (lsb)	130	136	200
Gate (ns)	400	390	150
Pre-Gate	20	14	4

III. ANALYSIS AND DISCUSSION

Through the coincidence of the detector with the Plastic Scintillator Detector the study of cosmic muons passing through the detector was possible.

^{241}Am α source was used to study the incidence position of the scintillator bars and demonstrated through FIG. 4. Along with the α energy deposition in the calorimeter FIG. 5. From the successful demonstration of incidence position and energy deposition, the detector is characterised.

For experimental procedures the reduction of background cosmic muons is important so it is necessary to study the behaviour of the detector setup to muons. The orientation of calorimeter is important, whether its parallel (horizontal) or perpendicular (vertical) to the Plastic Scintillator Detector. The settings used for coincidence are shown in Table I. These settings enabled the detection of vertical passing muons.

The energy deposited depends on the effective thickness that the muons pass through. The two different position configurations enable to examine two distinct effective thicknesses, as shown by comparing FIG. 9 with FIG. 8.

The observed behaviour demonstrates all the characteristics of the detector to be used for the Internal Pair Creation in ^{8}Be experiment at INFN in the future.

IV. CONCLUSION

Through the coincidence of the detector with the Plastic Scintillator Detector, it became possible to study cosmic muons passing through the detector. The ^{241}Am α source was utilized to investigate the incidence positions of the scintillator bars, as illustrated in FIG. 4, and the energy deposition in the calorimeter in horizontal and vertical configuration, as shown in FIG. 6 and FIG. 7. Although incidence positions and energy deposition were successfully demonstrated, the limited availability of only two data points prevented a complete calibration.

In experimental procedures, reducing background cosmic muon interference is important, which requires extensive research on the behavior of the detector towards muons. The orientation of the calorimeter to the Plastic Scintillator Detector significantly influences the results.

Nevertheless calibration was limited to two points, the observed behavior confirms that the detector exhibits all the necessary characteristics for future use in the Internal Pair Creation experiment with ^{8}Be.

[1] A. Krasznahorkay, M. Csatlós, L. Csige, Z. Gácsi, J. Gulyás, M. Hunyadi, I. Kuti, B. Nyakó, L. Stuhl, J. Timár, T. Tornyi, Z. Vajta, T. Ketel, and A. Krasznahorkay, "Observation of anomalous internal pair creation in ^{8}be: A possible indication of a light, neutral boson," *Physical Review Letters*, vol. 116, no. 4, jan 2016. [Online]. Available: http://dx.doi.org/10.1103/PhysRevLett.116.042501

[2] T. Nomura and P. Sanyal, "Explaining atomki anomaly and muon $g-2$ in $u(1)_x$ extended flavour violating two higgs doublet model," *Journal of High Energy Physics*, vol. 2021, no. 232, may 2021. [Online]. Available: https://link.springer.com/article/10.1007/JHEP05(2021)232